Chiranjeev Kumar

Data Science Consultant, Generative AI | Production Agentic AI, RAG, MCP & LLM Evaluation

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LinkedIn | GitHub | Portfolio: chiranjeev-ai-engineer.vercel.app | Kaggle

SUMMARY

GenAI engineer with **~6 years** across software, ML and data science who takes **multi-agent and RAG systems from prototype to production**. Shipped **RFP automation, an NLQ-to-SQL agent, a six-agent AP invoice platform and an AI tracing platform** to production at Deloitte, and built the **guardrail and observability SDK** that makes such systems safe to run. Strong on the last mile: prompt and model versioning, LLM-judge guardrails, evaluation, tracing and incident handling. Delivered a GenAI session to **1,000+ practitioners**; **Top 20 of ~1,000 teams** in an agentic AI hackathon; **Claude Certified Developer Associate** and **NVIDIA Certified Professional, Agentic AI**.

CORE SKILLS

Agentic AI & LLMs: LangGraph, LangChain, Haystack, multi-agent orchestration, ReAct tool loops, MCP servers/clients, human-in-the-loop gates, prompt engineering, fine-tuning, RAG, hybrid and vector search

Evaluation, Safety & Observability: LLM-as-judge (hallucination, toxicity, jailbreak, relevancy), schema and tool-call guards, PII masking (Presidio, detect-secrets), golden sets, MLflow tracing, LangFuse, Azure Monitor / App Insights, Grafana

Cloud & Inference: Azure OpenAI, Azure AI Search, Azure Document Intelligence, Microsoft Foundry, Azure ML, Container Apps, Key Vault, Entra ID, AWS, NVIDIA DGX / NeMo

Engineering: Python, FastAPI, async SQLAlchemy, Pydantic, SQL, PySpark, React 19, TypeScript, PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, Terraform, GitHub Actions, pytest, Vitest

Vision & Documents: GPT vision models, OpenCV, PyMuPDF, scikit-image, SIFT / SSIM alignment, Kornia LoFTR

Platforms & AI-assisted dev: Dataiku (Flow, LLM Mesh, Knowledge Banks, Prompt Studio, plugins); Claude Code, GitHub Copilot, Cursor, Windsurf, n8n

EXPERIENCE

Deloitte — **Data Science Consultant, Generative AI** Bengaluru | Oct 2024 – Present

Production GenAI systems

- **AP Invoice Hold Resolution (live in production):** built a LangGraph orchestrator that routes held supplier invoices by hold code to **six specialist agents** (price/PO, unit price, quantity, missing GR, partial GR, approval gap) plus an **email agent**. GPT-4o reasons over PO, GR and invoice data and **streams a cited root-cause report over SSE**; analysts edit and send the drafted supplier email. React 19 + FastAPI, **29 REST/SSE endpoints**, JWT, prompt versioning, feedback loop, **80%+ coverage** target.
- **RFP automation (production):** architected extraction, chunking, vector indexing, Azure OpenAI generation and Azure AI Search for enterprise proposals, cutting turnaround from **~4 weeks to ~2 (~50%)** within a **~$1M**, four-use-case engagement with **$5M+** follow-on pipeline.
- **NLQ-to-SQL agent (production):** multi-agent system in Dataiku that lets business users query enterprise databases in plain English, with automated evaluation of SQL correctness and latency; **45% lower** end-to-end query latency.
- **AI tracing platform (production):** one trace_id carried across gateway, orchestrator and every worker, giving a **full MLflow waterfall per request** plus a **billing-grade usage ledger**.
- Owned the **production lifecycle** (model and prompt versioning, Azure Monitor / App Insights monitoring, incident escalation), improving GenAI app response stability by **30%**.

Platforms, SDKs and accelerators

- **Guardian & Observability SDK (v1.6.1):** one Python package that adds **16 guardrail and 10 observability features** to any LangChain / Azure OpenAI app **by configuration**. Five-step *guarded()* pipeline (extract, check, policy, route pass/warn/block, persist), LLM-judge checks via Microsoft Foundry, L1/L2 payload masking, **fail-open** design, per-request MLflow runs with tokens and cost, hourly p95 and burn-rate rollups; self-hosted MLflow in a private Azure VNet with keyless Entra auth.
- **Vendor print drawing diff pipeline: nine-agent** Python backend that compares two revisions of engineering drawings (P&IDs, vendor prints). Combines Azure Document Intelligence OCR with GPT vision, runs **text and geometric (SIFT/SSIM) diffs in parallel**, validates each candidate in a **ReAct loop** and returns confidence-scored annotated PDFs through an async FastAPI.
- **AI change impact analysis:** designed the AI-assisted Trend flow from vendor print to updated Revit model (compare drawings, early scope/cost/schedule read, trace into Revit, pre-fill Trend Form), targeting **weeks-to-days** analysis with engineers keeping the go/no-go.
- **Ascend Agentic Harness:** contributed to a Claude Code delivery framework with **three phases, nine skills**, subagent TDD builds and deterministic quality gates, plus a **GitHub Copilot path** for teams without Claude Code access.

- **Copilot Core SDK:** async, multi-tenant agent engine (observe, plan, execute, validate loop with a verification clause, dependency-graph scheduler for **~2x faster** parallel steps), deployed as an AI copilot inside Autodesk Revit for **500+ developers**.
- Led **GenAI Centre of Excellence** incubator work: **10+ POCs** and production pipelines for RAG, multi-agent frameworks and MCP on Azure, NVIDIA DGX and AWS, raising operational efficiency by **25%**.

Enablement and recognition

- Delivered a GenAI session to **1,000+ practitioners** (**Applause Award, Oct 2026**), alongside earlier multi-agent and RAG trainings.
- **Agentic AI Hackathon: Top 20 of ~1,000 teams** (two-person team); Outstanding Performance Award submitted.
- **Applause Award 2026**, **Outstanding Performance Award 2025**, **Best Performer Award 2025**, **1st prize (team)** Consultant-Level Bangalore Hackathon.

Tredence Inc. — **Data Scientist** Bengaluru | May 2023 – Oct 2024

- Built an end-to-end **item-matching system** for a supermarket chain using LLM extraction: **95% accuracy across 50K+ SKUs**; manual matching time per item down **~60%** (~10 to ~4 minutes).
- Designed **RAG pipelines** and iterated prompts for product-attribute extraction, improving retrieval precision by **35%**.
- Benchmarked Dataiku, Databricks and Vertex AI on four metrics to drive platform selection, cutting projected infrastructure cost by **20%**.
- Productionised cloud-GPU inference against latency and cost SLAs, reducing cloud spend by **15%**. **Pat on the Back Award** for catalog normalization.

Soroco — **Machine Learning Software Engineer** Bengaluru | Dec 2021 – Apr 2023

- Built ML models and feature pipelines for large-scale behavioural data in enterprise process mining, with A/B testing frameworks that lifted model conversion by **12%**.

Tata Consultancy Services — **Associate Software Developer** Bengaluru | Nov 2020 – Dec 2021

- Provisioned Azure App Services and REST APIs; built ETL and data validation for enterprise ingest pipelines processing **1TB+ daily** at **99.9% uptime**.

SELECTED INDEPENDENT PROJECTS

- **Shikshak Sahayak:** Hindi-first AI assistant for Bihar Board teachers (lesson plans, assessments, textbooks, notices); **built and deployed in a weekend** with React, FastAPI and PostgreSQL.
- **Arogya Clinic:** production digital prescription platform for a general physician (React 19, FastAPI, PostgreSQL, bcrypt auth, drug catalog search, print view matching the paper format); co-founder of Arogya Sahayak.
- **Physical AI:** imitation learning and policy training with LeRobot on SO-100 and Koch v1.1 robotic arms; NVIDIA Isaac Sim for sim-to-real.

More projects and live demos: chiranjeev-ai-engineer.vercel.app

EDUCATION

Liverpool John Moores University
M.Sc., Machine Learning & AI
Distinction (73%) | 2022 – 2023

IIIT Bangalore
PG Diploma, Machine Learning & AI
3.5 / 4.0 | 2021 – 2022

BIT Mesra
B.E., Computer Science & Engineering
7.8 / 10 | 2016 – 2020

Publication: "TLCoV – An automated Covid-19 screening model", Elsevier (ScienceDirect), 2021

CERTIFICATIONS

- **Anthropic:** Claude Certified Developer Associate; Claude Certified Architect & Associate – Foundations (2026)
- **NVIDIA:** Certified Professional – Agentic AI (Dec 2025); Associate – Generative AI LLMs (Dec 2024); Associate – Generative AI Multimodal (Dec 2024)
- **Google:** Generative AI for Leaders (Jan 2026)
- **Microsoft:** Azure Data Scientist Associate (May 2024)
- **Coursera:** Prompt Engineering Specialization (Feb 2024)
- **Dataiku:** Core Designer; GenAI Practitioner
- **Kaggle:** Notebooks Expert, top 0.5% (rank 1,029 of 225K)
- All credentials: LinkedIn certifications